Homework Week 9

Modern Physics

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For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

ATOMS, RELATIVITY

- **Electronic ground states / Ex. 8.2**
  For all elements in the first two *periods*, list the electron descriptions in their ground state using $n\ell$ notation (for example, helium is $1s^2$).

- **Shells & subshells / Ex. 8.3**
  How many subshells are in the following shells: K, L, N, M, and O ($n = 1, \dots, 5$)?

- **Elements with no ground-state spin-orbit splitting / Ex. 8.14**
  List all the elements through calcium (Ca) that you would expect not to have a spin–orbit interaction that splits the ground-state energy. Explain.

  Hint: spin–orbit interaction is caused by a term proportional to $\mathbf{L} \cdot \mathbf{S}$*, where* $\mathbf{L}$ *and* $\mathbf{S}$ *correspond to the total orbital and spin angular momentum. When does this term vanish?*

- **Time dilation / Ex. 2.23**
  A clock in a spaceship moving at a constant velocity is observed to run at a speed of only 3/5 that of a similar clock at rest on Earth. How fast is the spaceship moving?

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